

Anurag Gupta

Senior Software Engineer | Data Engineering & Applied AI/LLM Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 17+ years of enterprise engineering experience, specialising in large-scale Data Engineering and Applied AI/LLM systems. Proven builder of production-grade data pipelines, cloud-native analytics platforms, and AI-powered automation on AWS and Databricks — delivering at enterprise scale in regulated financial services environments.

PROFESSIONAL EXPERIENCE

Senior Software Engineer, JPMorgan Chase

06/2022 – Present

Platform & Infrastructure:

London, UK

- Led modernisation of legacy on-prem data warehouse into a cloud-native Databricks analytics platform supporting large-scale workforce data across the firm.
- Built scalable data pipelines using Python, PySpark, Databricks, and AWS (S3, Glue, Lambda, ECS) processing enterprise-scale analytical workloads.
- Implemented prompt engineering strategies and output validation patterns to reduce hallucination risk in production financial systems.
- Implemented Infrastructure as Code with Terraform and CI/CD automation (Spinnaker) to standardise deployments and reduce operational risk.

AI Project: Automated Regression Suite (AWS Bedrock · Claude Sonnet · Pydantic · Great Expectations · Databricks)

- Conceived and independently built a production-grade AI-powered regression testing framework — the first of its kind within the team — autonomously validating data quality across 100+ production tables in a high-frequency deployment environment.
- Engineered an LLM-driven metadata analysis pipeline where Claude Sonnet autonomously identifies business-critical columns per table — replacing hardcoded rules with context-aware, AI-driven column prioritisation.
- Designed a dynamic validation code generation system where Claude Sonnet produces bespoke validation logic per table — eliminating manual test scripting entirely.
- Implemented structured output via Pydantic to capture all LLM-generated outputs as validated JSON before downstream processing — ensuring production reliability.
- Built end-to-end automated reporting pipeline ingesting Pydantic-structured results into Great Expectations, producing HTML reports consumed directly by Product Owners.

Lead Software Engineer, JPMorgan Chase

04/2015 – 04/2022

London, UK

- Led design and delivery of enterprise MIS and firmwide BI reporting solutions supporting executive decision-making across multiple business lines.
- Built Python-based data pipelines consuming REST API endpoints, applying business logic transformations, and persisting to target databases — in a heavily regulated financial environment with strict audit and compliance requirements.

- Translated complex business requirements into robust data models and reporting pipelines; owned full development lifecycle from design through production deployment.
- Served as primary technical point of contact for business stakeholders, translating enterprise user requirements into scalable engineering solutions and communicating technical decisions to non-technical audiences.
- Mentored junior engineers and coordinated cross-functional delivery across data, technology, and business stakeholder teams.

EDUCATION

MSc Digital & Technology Solutions (Data Science & Machine Learning, L7),
University of Exeter, UK

2025 – Present
UK

- Coursework: Machine Learning, NLP, Model Evaluation, AI Systems Design

Bachelor of Engineering, Electrical & Electronics Engineering – 1st Class with Distinction, *Visvesvaraya Technological University, Bangalore*

Bangalore, India

TECHNICAL SKILLS

Data Engineering Databricks, PySpark, Apache Spark, AWS Glue, AWS S3, AWS Lambda, ETL/ELT Pipelines, Data Modelling, Data Quality Frameworks, Great Expectations, SQL at Scale	Agentic AI LangGraph, ReAct Pattern, Multi-Agent Orchestration, Tool Use, Agent Memory & State
RAG & NLP Retrieval-Augmented Generation, ChromaDB, OpenSearch, Embedding-based Semantic Search, Document Indexing, NLP Pipelines, Text Classification, Named Entity Recognition	LLM & GenAI LangChain, OpenAI API (GPT-4o, GPT-4o-mini), AWS Bedrock (Claude), Prompt Engineering, Guardrails, LLM-as-Judge Evaluation, Knowledge Graph Integration
Cloud & MLOps Docker, AWS (S3, Glue, Lambda, EC2, ECS), Databricks, Terraform, CI/CD (Spinnaker), Streamlit, LangSmith	Python Production-grade Python, FastAPI, Async Python, Pydantic, PySpark, Pytest, scikit-learn
Data & ML Pandas, NumPy, TensorFlow, Supervised Learning, Feature Engineering, Model Evaluation	Other SQL, PL/SQL, Unix Shell, REST API Design, Git, Jupyter

AI ENGINEERING PROJECTS

FinSight, *AI-Powered Financial Research Platform* | *LangChain · LangGraph · ReAct · ChromaDB · OpenAI GPT-4o · FastAPI · Docker · LangSmith*

Phase 1 — RAG Pipeline:

- Built complete ingestion pipeline: PDF extraction (pdfplumber), recursive character chunking (1000 chars / 200 overlap), OpenAI embedding generation, and ChromaDB vector storage.
- Key architectural discovery: switched from PyPDFLoader to pdfplumber after identifying character corruption — reinforcing a data quality over model quality principle critical in production AI systems.
- Implemented semantic search retrieval with metadata filtering, returning top-K relevant chunks with relevance scoring.
- Designed prompt engineering layer with XML-tagged context, financial analyst system prompt, and hallucination prevention guardrails.

- Implemented dual-layer security: input guards (prompt injection detection) and output guards (PII / system prompt leakage prevention).
- Built interactive Streamlit chat interface with session-based conversation history and real-time response display.
- Integrated LangSmith observability — every query automatically traced with latency, token usage, and cost per request with zero code instrumentation.

Phase 2 — Autonomous Agent:

- Evolved RAG into a fully autonomous agent using LangGraph with the ReAct pattern — enabling iterative plan, tool selection, observation, and self-correction loops.
- Designed multi-tool agentic framework with four specialised tools: SEC filing retrieval (RAG), financial data lookup, web search integration, and calculation engine — invoked autonomously based on query context.
- Implemented LangGraph state management for agent memory, multi-step reasoning loops, and conditional tool routing across complex financial queries.
- Developed automated evaluation pipeline across 12 test cases at 4 difficulty levels: keyword fact-matching (60.15%) and LLM-as-judge scoring (72.22%); achieved 100% accuracy on unanswerable questions.
- Containerised full stack with Docker and docker-compose; exposed RESTful API via FastAPI with Pydantic validation and API key authentication. Average query latency: 2.95 seconds end-to-end.

ConvFinQA,

Conversational Financial QA Agent | LangGraph · GPT-4o-mini · Tool Calling · Prompt Engineering · Python

- Built a conversational agent for multi-hop numerical reasoning over financial documents using LangGraph with stateful conversation history — enabling Q2 to reference Q1 answers across chained questions.
- Implemented tool-augmented arithmetic via five calculator functions bound to GPT-4o-mini at temperature 0 — arithmetic never computed in the model's head.
- Achieved 70.8% accuracy across 185 questions on the ConvFinQA benchmark through three systematic prompt iterations — few-shot examples outperformed rule-based instructions (60.5%).
- Conducted systematic failure analysis identifying three distinct error categories: division order errors, percentage share vs change confusion, and conversational pronoun resolution failures.
- Implemented dual-metric evaluation: exact match for lookup questions and numerical accuracy within 1% tolerance for calculated answers.

CERTIFICATIONS

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| • AWS Certified Cloud Practitioner | • AWS Certified Solutions Architect |
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